Department of B. Tech First Year
I[st] Sem Session 2026-27 (ODD Semester)
Assignment – 1 & 2

Subject Name: Introduction to AI and Prompt Engineering **Subject Code: AAI 101**

Date of Assignment: 28.10.2026 **Date of Submission: 05.11.2026**

Q.No.	Questions	Marks
1	Concept-check task. Pick one real ML-based system relevant to your branch (e.g., a recommendation engine, a defect detector, a chatbot). Identify: (a) Is it supervised, unsupervised, or reinforcement learning? (b) What could cause it to overfit or underfit? (c) Sketch a simple diagram showing its inputs, a basic neuron/layer concept, and output. Submit as a half-page note (200–250 words) with diagram.	5
2	**i).** What is centroid in K- Means? **ii).** What is MAE? Where is it used? **iii).** Name the three types of layers in a neural network.	6
3	Compare AI, ML, DL with suitable diagram and examples	5
4	Discuss the various real-life applications of Artificial Intelligence in Google Maps, Netflix, Spotify, Google Photos and voice assistants.	5
5	Explain the k -means clustering algorithm step by step with an example.	5
6	Explain the confusion matrix and the metrics accuracy, precision, recall and F1 score with a numerical example.	5
7	Explain NLP, Its Stages and applications.	5
8	A neuron has inputs x = (2, 3), weights w = (0.5, −0.4) and bias 0.2. Find its output with sigmoid and with ReLU activation.	3

Signature & Name of Faculty **HoD Signature**

Date: **Date:**

Note: Use pages to write the answers. Avoid copying assignments